

KYC Verification Policy

Policy and decision rules for ID information extraction, face verification, and video liveness assessment.

1. Purpose

This policy defines the decision rules used by the KYC verification system. The system evaluates Photo ID information, face verification evidence, and video liveness evidence. The policy supports three outcomes: PASS, REVIEW, and FAIL.

2. ID Information Requirements

The required Photo ID fields are Name, Date of Birth (DOB), and ID Number.

Condition	Decision
Required ID fields are successfully extracted.	Continue with verification.
One or more required ID fields cannot be extracted.	REVIEW

3. Face Verification

Face verification compares the face embedding generated from the Photo ID with a usable face embedding obtained from the KYC video. A valid comparison requires a single detectable face in the relevant input.

Condition	Decision
No face detected in the KYC video.	FAIL
More than one face is detected in the KYC video.	FAIL
No single face is detected in the Photo ID.	FAIL
No usable single-face embedding can be obtained from the KYC video.	FAIL
Face distance is below 0.50.	PASS
Face distance is 0.50 or higher.	REVIEW

4. Liveness Verification

The video liveness check uses detected blink activity and lip movement activity. Audio is not required for lip movement detection; the assessment is based on facial landmark movement in the video.

Condition	Decision
Blink count is greater than 0 AND lip movement count is greater than 0.	PASS
Blink count is 0 OR lip movement count is 0.	REVIEW
No face is detected in the video.	FAIL
Multiple faces are detected in the video.	FAIL

5. Overall Decision Rules

The following rules define how the evidence is classified. A FAIL condition is conclusive for the corresponding verification requirement. REVIEW is used where the evidence is insufficient or falls outside the automatic PASS condition.

Evidence / condition	Outcome
Required ID field(s) cannot be extracted.	REVIEW
No face detected in video.	FAIL
Multiple faces detected in video.	FAIL
No single face detected in Photo ID.	FAIL
No usable video face embedding.	FAIL
Face distance < 0.50.	PASS
Face distance >= 0.50.	REVIEW
Blink count > 0 AND lip movement count > 0.	PASS
Blink count = 0 OR lip movement count = 0.	REVIEW

6. Outcome Definitions

Outcome	Meaning
PASS	The relevant evidence satisfies the automatic pass condition.
REVIEW	The evidence is inconclusive, incomplete, or outside the automatic pass condition and requires further review.
FAIL	A conclusive failure condition has been detected.

7. Example Evidence

Example A: Face distance = 0.41, blink count = 1, lip movement count = 1. The face comparison satisfies the PASS threshold and both liveness indicators are present.

Example B: Face distance = 0.72, blink count = 1, lip movement count = 1. The face comparison is REVIEW because the distance is not below 0.50, even though liveness indicators are present.

Example C: Face distance = 0.41, blink count = 0, lip movement count = 1. The face comparison satisfies PASS, but liveness is REVIEW because blink evidence is absent.

Example D: Multiple faces are detected in the video. The liveness result is FAIL and the video does not satisfy the single-face requirement.

Example E: No single face is detected in the Photo ID. Face verification is FAIL because the Photo ID face is required for comparison.

Example F: A required ID field cannot be extracted. The ID information is incomplete and the case is REVIEW.

8. Scope of This Policy

This document defines the verification rules and evidence classifications used by the KYC system. It is intended to serve as a policy reference for retrieving the relevant rules when evaluating KYC evidence.